

Andrew Bai

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EXPERIENCE

Onboard Payload Software Co-op

May 2026 - August 2026

Telesat

Ottawa, Ontario

- Shipped production code in C++ and wrote component level tests in Python for Telesat's low earth orbit satellite constellation.
- Supported development of message intake logic, MODCOD control loop, and debug command line interface, as well as many other general maintenance/debugging tasks.
- Collaborated in an Agile/Scrum environment, managed tasks and merge requests in Jira and GitLab, including CI/CD pipeline usage.

General Autonomy Member

January 2026 – Present

Waterloo Aerial Robotics Group

Waterloo, Ontario

- Participated as a general autonomy member for the group's competition drone.
- Implemented HSV-based color detection in Python and Open CV
- Worked on a multi-process drone ground station in Python using MAVLink messaging and inter process queues to communicate with and control a simulated drone.

PROJECTS

Hexapod Robot

April 2026 - July 2026

- Designed and 3D printed a fully custom six-legged hexapod robot from scratch in SolidWorks, modeling all mechanical components and leg linkages
- Programmed leg kinematics and gait control in C++ on an ESP32 microcontroller, achieving stable coordinated walking at speeds up to 0.5 km/h
- Designed circuit diagrams and wired the full electrical system by hand, including soldering all connections between the ESP32, servos, and power distribution

Automatic Poker Dealer

October 2025 - November 2025

- Collaborated with three teammates to design and build a poker dealing robot.
- Programmed game logic, sensor inputs, and card dispensing in C++ using the VEX IDE.
- Integrated a color sensor to intake bets, buttons to progress the game, and a distance sensor to ensure consistent card dealing.

Smart Knee Brace

February 2026

- Built a smart knee brace which adjusts its support based on how vulnerable of a position the knee is in.
- An Arduino nano took in gyroscope, bend sensor, and pressure sensor data to choose the most optimal position for a servo.
- Placed 3rd among 264 participants at makeUofT

EDUCATION

University of Waterloo

Waterloo, Ontario

Bachelor of Applied Science in Mechatronics Engineering, GPA: 85%

September 2025 - June 2030

- **Relevant Coursework:** Dynamics, Circuit Analysis, Data Structures & Algorithms, Numerical Methods, Material Science

TECHNICAL SKILLS

Programming & Software: Python, C++, JavaScript, React, pandas, NumPy, Matplotlib, OpenCV

Embedded Systems & Hardware: ESP32, Arduino, circuit design, Soldering, SolidWorks, 3D printing

Developer Tools: Git, GitLab, Docker, VS Code, Jira, CI/CD pipelines, Agile/Scrum, component-level testing, LaTeX